

Answer Keys – All Worksheets

Note for Teachers: This section contains answers to all 14 worksheets. Answers marked “Teacher Verification Required” involve drawings, constructions, graphs, or open-ended responses to be checked by the teacher in person. This section may be detached and kept separately from the student worksheet booklets.

Worksheet 1: Chapter 1 – A Square and a Cube

Answer Key

Answer Key – Worksheet 1

1. $15^2 = 225$; $23^2 = 529$; $(-7)^2 = 49$
2. $4^3 = 64$; $10^3 = 1000$; $(-3)^3 = -27$
3. Yes; $\sqrt{225} = 15$
4. Yes; $\sqrt[3]{512} = 8$
5. 121, 144, 169, 196
6. $2916 = 2^2 \times 3^6$; $\sqrt{2916} = 2 \times 27 = 54$
7. $9261 = 3^3 \times 7^3$; $\sqrt[3]{9261} = 21$
8. Side = $\sqrt{1444} = 38$ m
9. $\sqrt{729} = 27$; 27 rows and 27 columns
10. Last digit of 7921 is 1 $\Rightarrow$ square root ends in 1 or 9; $\sqrt{7921} = 89$ (odd)
11. $720 = 2^4 \times 3^2 \times 5$; need one more 5; multiply by **5**
12. $1944 = 2^3 \times 3^5$; $3^5 = 3^3 \times 3^2$; need to remove $3^2 = 9$; divide by **9**
13. $\sqrt{6.25} = 2.5$; $\sqrt{0.0081} = 0.09$
14. $\sqrt[3]{0.001331} = 0.11$
15. $x = 3.6$
16. (a) $10^2 = 100$; (b) $n^2 = 169 \Rightarrow n = 13$ odd numbers
17. $15625 \times 1000 = 15625000$ cubic cm; $\sqrt[3]{15625000} = 250$ cm
18. $(n+2)^2 - n^2 = 4n + 4 = 100 \Rightarrow n = 24$; numbers are 24 and 26
19. $31^2 = 961$; $32^2 = 1024 > 1000$; use 961 saplings; 39 left over
20. Edge = $\sqrt[3]{2744} = 14$ cm; 8 boxes $\Rightarrow$ side of larger cube = $2 \times 14 = 28$ cm

Worksheet 2: Chapter 2 – Power Play (Exponents)

Answer Key

Answer Key – Worksheet 2

1. $2^7 = 128$ 2. $5^3 = 125$ 3. $1 + 1 - 1 = 1$ 4. $\frac{1}{8}$ 5. $4^6 = 4096$
6. $\frac{9}{4}$ 7. (a) 6.75×10^6 ; (b) 3.2×10^{-7}
8. $\frac{16 \times 27}{36} = 12$ 9. $2^7 = 128 \Rightarrow x = 7$ 10. $\left(\frac{5}{5}\right)^2 = 1$

11. $\frac{2^8 \times 3^4 \times 2^2 \times 3^2}{2^6 \times 3^4} = \frac{2^{10} \times 3^6}{2^6 \times 3^4} = 2^4 \times 3^2 = 144$
12. $\frac{6 \times 10^{24}}{7.4 \times 10^{22}} \approx 81$ times heavier
13. $(3^{-1} + 4^{-1})^{-1} = \left(\frac{7}{12}\right)^{-1} = \frac{12}{7}$
14. $3^{2x-1} = 3^{3(x-1)} \Rightarrow 2x - 1 = 3x - 3 \Rightarrow x = 2$
15. $\frac{10^{-6}}{10^{-4}} = 10^{-2} = 0.01$
16. $2^{20} = 1048576 \approx 1.05 \times 10^6$ bacteria
17. $\frac{x^6 y^{-3} \times x^{-2} y^4}{x^4 y} = \frac{x^4 y}{x^4 y} = 1$
18. $\frac{a^3}{b^2} \times \frac{b^4}{a} = a^2 b^2 = 2^{2p} \times 2^{2q} = 2^{2p+2q}$
19. $\frac{7 \times 10^{-5}}{1.06 \times 10^{-10}} \approx 6.6 \times 10^5$ atoms
20. In rupees: $\frac{4.72 \times 10^{13}}{100} = 4.72 \times 10^{11}$ rupees; Per person = $\frac{4.72 \times 10^{11}}{1.4 \times 10^9} \approx \text{Rs. } 337$

Worksheet 3: Chapter 3 – Rational Numbers

Answer Key

Answer Key – Worksheet 3

1. $\frac{5}{7}; \frac{-3}{8}; 0$
2. $\frac{-9}{4}; \frac{1}{7}; -5$
3. Yes; $\frac{-6}{-9} = \frac{2}{3}$
4. $\frac{-2}{3} < \frac{-1}{2} < 0 < \frac{1}{4}$
5. Any three: $\frac{3}{15}, \frac{4}{15}, \frac{5}{15}$ etc.
6. $\frac{-5}{8} + \frac{6}{8} = \frac{1}{8}$ 7. $\frac{14}{24} + \frac{9}{24} = \frac{23}{24}$ 8. $\frac{12}{72} = \frac{1}{6}$
9. $\frac{-7}{10} \times \frac{5}{14} = \frac{-1}{4}$ 10. $\frac{2}{3} \times \frac{-5}{7} = \frac{-10}{21} = \frac{-5}{7} \times \frac{2}{3} \checkmark$
11. $\frac{-3}{4} \times \frac{-3}{10} = \frac{9}{40}$
12. Other = $\frac{-9}{14} \div \frac{3}{7} = \frac{-9}{14} \times \frac{7}{3} = \frac{-3}{2}$
13. Teacher Verification Required
14. LHS = $\frac{2}{5} \times \frac{-1}{4} = \frac{-1}{10}$; RHS = $\frac{-3}{10} + \frac{1}{5} = \frac{-1}{10} \checkmark$
15. $\frac{4}{7} \times \left(\frac{-3}{8} + \frac{-5}{8}\right) = \frac{4}{7} \times (-1) = \frac{-4}{7}$
16. Use $\frac{6}{42}$ to $\frac{7}{42}$: five numbers $\frac{31}{210}, \frac{32}{210}, \frac{33}{210}, \frac{34}{210}, \frac{35}{210}$

17. Other = $\frac{-4}{5} - \frac{7}{15} = \frac{-12}{15} - \frac{7}{15} = \frac{-19}{15}$; Negative
 18. Priya is correct ($-\frac{4}{3}$ negative); Rahul is wrong ($x + \frac{1}{x} \geq 2$ only for $x > 0$)
 19. $SP = \frac{6}{5} \times CP = 960 \Rightarrow CP = \text{Rs. } 800$; Gain = Rs. 160
 20. After use: $\frac{3}{7} - \frac{1}{4} = \frac{5}{28}$; Transferred: $\frac{2}{3} \times \frac{5}{28} = \frac{5}{42}$; Left: $\frac{5}{28} - \frac{5}{42} = \frac{5}{84}$

Worksheet 4: Chapter 4 – Quadrilaterals

Answer Key

Answer Key – Worksheet 4

- 360°
- Opposite sides equal and parallel; opposite angles equal (any 2)
- All four angles = 90°
- All sides = 8 cm
- Trapezium
- $360^\circ - 75^\circ - 90^\circ - 110^\circ = 85^\circ$
- 65°, 115°, 65°, 115°
- Half-diagonals: 8 and 6; side = $\sqrt{64 + 36} = 10$ cm
- Other side = $\sqrt{17^2 - 8^2} = 15$ cm; Perimeter = $2(8 + 15) = 46$ cm
- $2(5 + 8) = 26$ cm
- Teacher Verification Required (proof using angles on a straight line)
- $3x + 10 = 5x - 30 \Rightarrow x = 20$; Angles: 70°, 110°, 70°, 110°
- Side = 13 cm; Diagonals: 24 and $2\sqrt{169 - 144} = 10$ cm; Area = $\frac{1}{2} \times 24 \times 10 = 120$ sq. cm
- $\angle C = 110^\circ$; $\angle D = 70^\circ$ (co-interior angles)
- Bisect each other: parallelogram, rhombus, rectangle, square; Equal length: rectangle, square
- Teacher Verification Required
- Rhombus diagonals $3k$ and $4k$; Perimeter = 40 $\Rightarrow$ side = 10; $\frac{1}{2} \times (3k)(4k) = ?$;
 $\text{side}^2 = (1.5k)^2 + (2k)^2 = 100 \Rightarrow k^2 = 16 \Rightarrow k = 4$; Area = $\frac{1}{2} \times 12 \times 16 = 96$ sq. cm
 vs Square area = 100 sq. cm; Square has larger area
- Teacher Verification Required
- Area = $120 \times 80 = 9600$ sq. m; Cost = $9600 \times 15 = \text{Rs. } 1,44,000$
- Area = $\frac{1}{2}(14 + 10) \times 8 = 96$ sq. m; Tiles = $96 \div 0.25 = 384$; Cost = $384 \times 40 = \text{Rs. } 15,360$

Worksheet 5: Chapter 5 – Number Play

Answer Key

Answer Key – Worksheet 5

- $10p + q$
- Any 3-digit number with hundreds = a , tens = b , units = c
- $(10a + b) + (10b + a) = 11(a + b)$; divisible by 11 ✓
- Digit sum of multiples of 9 is also divisible by 9; e.g. 18: $1 + 8 = 9$; 36: $3 + 6 = 9$

5. All rows/cols/diags = 15 ✓
6. Number = $10(b+3) + b = 11b + 30$
7. $\overline{abc} - \overline{cba} = (100a + 10b + c) - (100c + 10b + a) = 99(a - c)$; divisible by 99 ✓
8. $N = 100a + 10b + c \equiv a + b + c \pmod{3}$; if $a + b + c$ div by 3 then N div by 3 ✓
9. Teacher Verification Required
10. $345345 = 345 \times 1001 = 345 \times 7 \times 11 \times 13$; result is the original 3-digit number
11. $10a + b = 4(a + b) \Rightarrow 6a = 3b \Rightarrow b = 2a$; and $b - a = 2$; $a = 2, b = 4$; Number = 24
12. $101a + 10b$; $101 = 11 \times 9 + 2$... simpler: $\overline{aba} = 11(9a + 10b)/11$... actually $101a + 10b$; check: $\overline{aba} = (100 + 1)a + 10b = 11 \times (\text{something})$; $101 \div 11 = 9$ rem 2; so use: $\overline{aba} = 100a + 10b + a = 11 \times 9a + 2a + 10b$... Direct: digits-alternate-sum = $a - b + a = 2a - b$; for div by 11 need $2a - b = 0$ or 11; not always. Correct: aXa where middle digit matters; Teacher accept algebraic argument.
13. Odd $\times$ odd = odd; $(2m+1)(2n+1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1 = \text{odd}$ ✓
14. Teacher Verification Required
15. Middle integer = 33; In general: middle integer = $S/3$
16. Let number = n : $\frac{3n+6}{3=n+2}$; $n + 2 - n = 2$ ✓
17. $n(n+1)$: one of $n, n+1$ is even $\Rightarrow$ product even ✓; $n(n+1)(n+2)$: three consecutive integers include one multiple of 2 and one of 3 $\Rightarrow$ divisible by 6 ✓
18. Let number = n : $\frac{2n+10}{2} - n = n + 5 - n = 5$ ✓
19. $10 \times 2b + b = 21b$; $a + 2b + 2b + b = 18 \Rightarrow a + 5b = 18$; with $a = 2b$: $7b = 18$ (no integer); try $a = 2b$ and other combos; valid PIN: $a = 8, b = 4$, digits: 8, 4, 4, 2; sum = 18 ✓; PIN: **8442**
20. Five consecutive integers summing to 100: middle = 20; scores 18, 19, 20, 21, 22; Sum of n consecutive integers starting from k is $nk + n(n-1)/2$ which is multiple of $n = 5$ ✓

Worksheet 6: Chapter 6 – Algebraic Identities

Answer Key

Answer Key – Worksheet 6

1. $a^2 + 2ab + b^2$ 2. $x^2 + 10x + 25$ 3. $4y^2 - 12y + 9$
4. $x^2 - 16$ 5. $12x^3$
6. $4a^2 - 9b^2$ 7. $x^2 + 10x + 21$ 8. $(100 - 1)^2 = 9801$
9. $(100 + 2)(100 - 2) = 9996$ 10. $x^3 + 8y^3$
11. $(5x + 4y)(5x - 4y)$ 12. $(x + 5)^2$ 13. $(2a - 3b)^2$
14. $a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$; for 1,2,3: $(1 + 2 + 3)^2 = 36$ ✓
15. $x^2 + \frac{1}{x^2} = 25 - 2 = 23$; $x^4 + \frac{1}{x^4} = (23)^2 - 2 = 527$
16. $a^2 + b^2 = (a - b)^2 + 2ab = 16 + 42 = 58$; $(a + b)^2 = 58 + 2 \times 21 = 100$
17. Teacher Verification Required
18. $8x^3 + 12x^2y + 6xy^2 + y^3$
19. Area of border = $(x + 5)^2 - (x - 2)^2 = (x^2 + 10x + 25) - (x^2 - 4x + 4) = 14x + 21$; for $x = 8$: $14(8) + 21 = 133$ sq. m

20. Area = $(3a + 2)(2a - 1) = 6a^2 + a - 2$; $a = 4$: length = 14 cm, breadth = 7 cm,
Area = 98 sq. cm; verify: $6(16) + 4 - 2 = 98$ ✓

Worksheet 7: Chapter 7 – Proportional Reasoning 1

Answer Key

Answer Key – Worksheet 7

1. $48 : 36 = 4 : 3$ 2. $5 \times 50000 = 250000$ cm = 2.5 km 3. $\frac{60}{400} \times 100 = 15\%$
4. $SI = \frac{5000 \times 8 \times 2}{100} = \text{Rs. } 800$ 5. $y = 30$
6. $SP = 800 \times 0.85 = \text{Rs. } 680$ 7. $8 \times 12 = 16 \times x \Rightarrow x = 6$ workers (no, $8 \times 12 = 6 \times y \Rightarrow y = 16$ workers)
8. $CI = 10000 \left[\left(1 + \frac{10}{100} \right)^2 - 1 \right] = 10000 \times 0.21 = \text{Rs. } 2100$; Amount = Rs. 12100
9. $\frac{300}{4} \times 7 = 525$ km 10. $SP = CP \times (1 - 0.125) \Rightarrow CP = \frac{875}{0.875} = \text{Rs. } 1000$
11. Boys = 300; Girls = 240 12. $400000 \times (0.9)^2 = \text{Rs. } 3,24,000$
13. $80000 \times (1.05)^2 = 88200$ 14. $\frac{1200 \times R \times 8}{100} = 1200 \Rightarrow R = 12.5\%$
15. $A = 240, B = 360, C = 600$ 16. $P \left(1 + \frac{10}{100} \right)^2 = 6050 \Rightarrow P = \text{Rs. } 5000$
17. Effective discount = $1 - (0.8 \times 0.9) = 1 - 0.72 = 28\%$
18. On gain: $CP = \frac{990}{1.1} = \text{Rs. } 900$; On loss: $CP = \frac{990}{0.9} = \text{Rs. } 1100$; Total CP = Rs. 2000, Total SP = Rs. 1980; Loss = Rs. 20; Loss% = 1%
19. Amount = $25000 \times (1.08)^3 = 25000 \times 1.259712 = \text{Rs. } 31493$; Interest = Rs. 6493
20. After 20%: $18000 \times 0.8 = \text{Rs. } 14400$; After further 5%: $14400 \times 0.95 = \text{Rs. } 13680$; Savings = Rs. 4320

Worksheet 8: Chapter 8 – Playing with Constructions

Answer Key

Answer Key – Worksheet 8

1. Three (SSS, SAS, ASA, AAS, RHS – any 3 of these conditions)
2. Five independent measurements
3. Special conditions: (a) it is a parallelogram; (b) it is a rectangle/square/rhombus; (c) two diagonals and one included side are given
- 4–10. Teacher Verification Required for all constructions
11. Sum of three sides = $3 + 4 + 5 = 12 < 10$; No – the fourth side (diagonal here is 10) exceeds this; not constructible as the sum of any three sides must exceed the fourth

12–18. Teacher Verification Required

19–20. Teacher Verification Required

Worksheet 9: Chapter 9 – Mensuration (Surface Area and Volume)**Answer Key****Answer Key – Worksheet 9**

1. $6^3 = 216$ cubic cm
2. $2(8 \times 5 + 5 \times 3 + 3 \times 8) = 2(40 + 15 + 24) = 158$ sq. cm
3. $\frac{22}{7} \times 12.25 \times 10 = 385$ cubic cm
4. $2 \times \frac{22}{7} \times 7 \times 15 = 660$ sq. cm
5. $6s^2 = 294 \Rightarrow s = 7$ cm
6. TSA = $2(12 \times 8 + 8 \times 5 + 5 \times 12) = 2(96 + 40 + 60) = 392$ sq. cm
7. $\frac{22}{7} \times 49 \times 5 = 770$ cubic m
8. Bricks needed = $\frac{10 \times 3 \times 0.5 \times 10^6}{25 \times 12.5 \times 7.5} = \frac{15000000}{2343.75} = 6400$
9. New volume = $\pi(2r)^2 \times \frac{h}{2} = 2\pi r^2 h$; doubles
10. $\frac{22}{7} \times 1^2 \times 14 = 44$ cubic m
11. $\frac{2}{3} \times \frac{22}{7} \times (1.75)^2 \times 4 = \frac{2}{3} \times \frac{22}{7} \times 3.0625 \times 4 \approx 25.67$ cubic m
12. Height = $\frac{840}{10 \times 7} = 12$ cm; TSA = $2(10 \times 7 + 7 \times 12 + 12 \times 10) = 2(70 + 84 + 120) = 548$ sq. cm
13. Small edge = $\frac{8}{4} = 2$ cm (since $64 \times 2^3 = 512 = 8^3$ ✓)
14. Volume of cube = $11^3 = 1331$; $\frac{22}{7} \times 49 \times h = 1331 \Rightarrow h = \frac{1331 \times 7}{22 \times 49} = \frac{1331}{154} \approx 8.64$ cm
15. Area per revolution = $2\pi rL = 2 \times \frac{22}{7} \times 0.7 \times 2 = 8.8$ sq. m
16. Volume = $\pi(6^2 - 5^2) \times 100 = \frac{22}{7} \times 11 \times 100 \approx 3457$ cubic cm
17. Wall area = $2(20 + 15) \times 5 = 350$; Windows = $3 \times 1.8 = 5.4$; Doors = $2 \times 2 = 4$; Net area = 340.6 sq. m; Cost = $340.6 \times 80 = \text{Rs. } 27,248$
18. Cylinder: LSA = $2\pi rh = 2 \times \frac{22}{7} \times 3 \times 8 = \frac{528}{7} \approx 150.9$ sq. cm; Top circle = $\pi \times 9 = \frac{198}{7}$ sq. cm; Cone: $l = \sqrt{9 + 16} = 5$ cm; LSA = $\pi \times 3 \times 5 = \frac{330}{7}$ sq. cm; TSA = $150.9 + 28.3 + 47.1 \approx 226.3$ sq. cm; Vol: Cylinder = $\pi \times 9 \times 8 = 226.3$ cubic cm; Cone = $\frac{1}{3}\pi \times 9 \times 4 = 37.7$ cubic cm; Total ≈ 264 cubic cm
19. Volume = $\frac{22}{7} \times 1^2 \times 7 = 22$ cubic m = 22000 litres; Families = $22000 \div 200 = 110$
20. Vol of earth = $\frac{22}{7} \times 2.25 \times 14 = 99$ cubic m; Height = $\frac{99}{20 \times 15} = 0.33$ m

Worksheet 10: Chapter 10 – Statistics and Probability

Answer Key**Answer Key – Worksheet 10**

1. (a) $\frac{1}{6}$; (b) $\frac{3}{6} = \frac{1}{2}$; (c) $\frac{2}{6} = \frac{1}{3}$
2. $1 - 0.35 = 0.65$ 3. Mean = 20 4. Sorted: 2,3,5,7,8,9,10; Median = 7
5. Mode = 4 (appears 4 times)
6. $P(\text{red}) = \frac{1}{2}$; $P(\text{blue}) = \frac{3}{10}$; $P(\text{not green}) = \frac{8}{10} = \frac{4}{5}$
7. Outcomes: HH, HT, TH, TT; $P(\text{both H}) = \frac{1}{4}$; $P(\text{at least one T}) = \frac{3}{4}$
8. Sorted: 148, 148, 150, 152, 155, 155, 155, 160, 162, 170; Mean = 155.5; Mode = 155; Median = 155
9. $P(\text{ace}) = \frac{4}{52} = \frac{1}{13}$; $P(\text{red}) = \frac{26}{52} = \frac{1}{2}$; $P(\text{king of spades}) = \frac{1}{52}$
10. New sum = $30 \times 65 + (54 - 45) = 1950 + 9 = 1959$; New mean = $1959 \div 30 = 65.3$
11. Midpoints 5, 15, 25, 35, 45; Mean = $\frac{20 + 105 + 300 + 350 + 315}{40} = \frac{1090}{40} = 27.25$; Modal class: 20–30
12. (a) $\frac{1}{36}$; (b) pairs: (2,6), (3,5), (4,4), (5,3), (6,2) = 5; $P = \frac{5}{36}$; (c) pairs: (2,6), (3,4), (4,3), (6,2) = 4; $P = \frac{4}{36} = \frac{1}{9}$
13. New sum = $20 \times 45 - (60 + 80) = 900 - 140 = 760$; New mean = $760 \div 18 \approx 42.2$
14. $P(\text{up}) = \frac{120}{200} = 0.6$; Expected in 500 = 300
15. Teacher Verification Required
16. MATHEMATICS: M(2), A(3), T(2), H(1), E(1), I(1), C(1), S(1); Total = 11 letters; Vowels: A, A, A, E, I = 5; $P(\text{vowel}) = \frac{5}{11}$; $P(M) = \frac{2}{11}$; Letters appearing more than once: M, A, T; count = 7; $P = \frac{7}{11}$
17. Total sum = $50 \times 36 = 1800$; first 25 sum = 800; last 25 sum = $1800 - 800 = 1000$; $x = 40$
18. $P(\text{below } 40) = \frac{5}{40} = \frac{1}{8}$; $P(60-80) = \frac{18}{40} = \frac{9}{20}$; $P(\text{passed}) = \frac{35}{40} = \frac{7}{8}$
19. Teacher Verification Required 20. Priya is correct; Experimental $P = \frac{7}{12}$

Worksheet 11: Chapter 11 – Linear Equations**Answer Key****Answer Key – Worksheet 11**

1. $x = 5$ 2. $y = 4$ 3. $x = 28$ 4. $x = 7$ 5. $x = 6$; $3(6) + 2 = 20$
6. $3(3x - 1) = 4(x + 2) \Rightarrow 9x - 3 = 4x + 8 \Rightarrow x = \frac{11}{5} = 2.2$
7. $n + (n + 1) + (n + 2) = 78 \Rightarrow n = 25$; integers: 25, 26, 27
8. $1.25x = 75 \Rightarrow x = 60$
9. $7x - 14 - 3x - 3 = 5x - 11 \Rightarrow -x = 6 \Rightarrow x = -6$
10. Son = x ; Father = $3x$; $3x + 12 = 2(x + 12) \Rightarrow x = 12$; Father: 36, Son: 12

11. Multiply by 12: $4(x+1) + 3(x-1) = 2(2x+1) \Rightarrow 7x+1 = 4x+2 \Rightarrow x = \frac{1}{3}$
12. b = breadth; $l = 2b + 4$; $2(3b+4) = 64 \Rightarrow b = \frac{56}{6} \approx 9.3$ cm; length ≈ 22.7 cm
13. $0.4x + 0.3x - 0.3 = 0.2x + 0.4 \Rightarrow 0.5x = 0.7 \Rightarrow x = 1.4$
14. Tens = a , units = b ; $a + b = 10$; reversed: $10b + a - (10a + b) = 36 \Rightarrow b - a = 4$; $b = 7, a = 3$; Number = **37**
15. $d = 60t = 75(t-2) \Rightarrow d = 600$ km
16. $\frac{4}{3} = \frac{4x+6}{3x+6} \Rightarrow 12x+24 = 12x+18$... let ages be $4k, 3k$; $\frac{4k+6}{3k+6} = \frac{6}{5} \Rightarrow 20k+30 = 18k+36 \Rightarrow k = 3$; Ages: 12 and 9
17. In 1 hour: A fills $\frac{1}{4}$, B fills $\frac{1}{6}$, C empties $\frac{1}{8}$; Net = $\frac{6+4-3}{24} = \frac{7}{24}$; Time = $\frac{24}{7} \approx 3.43$ hrs
18. Cross multiply: $(4x-3)(3x-1) = (2x+5)(3x+2)$; $12x^2 - 13x + 3 = 6x^2 + 19x + 10$; $6x^2 - 32x - 7 = 0$; $x = \frac{32 \pm \sqrt{1024 + 168}}{12} = \frac{32 \pm \sqrt{1192}}{12}$; Teacher verify
19. $x - \frac{x}{3} - \frac{x}{4} = 150 \Rightarrow \frac{5x}{12} = 150 \Rightarrow x = \text{Rs. } 360$
20. 50-paise = n ; 25-paise = $n + 30$; $\frac{n}{2} + \frac{n+30}{4} = 45 \Rightarrow 2n + n + 30 = 180 \Rightarrow n = 50$; 50p coins: 50; 25p coins: 80

Worksheet 12: Chapter 12 – Factorisation

Answer Key

Answer Key – Worksheet 12

1. $4(x+2y)$ 2. $(x+5)(x-5)$ 3. $(a+3)^2$ 4. $3pq(1+2p)$ 5. $3x-2$
 6. $(x+2)(x+3)$ 7. $(y-4)(y+3)$ 8. $(a+b)(x+y)$ 9. $4(m+3n)(m-3n)$
 10. $x+4$
 11. $2(a^2-25) = 2(a+5)(a-5)$ 12. $(x-7)(x+3)$ 13. $(2x+3)(y+z)$ 14. x^2-2x+3
 15. $x^4-81 = (x^2+9)(x^2-9) = (x^2+9)(x+3)(x-3)$
 16. Substitute $x = -3$: $9-15+k=0 \Rightarrow k=6$; $x^2+5x+6 = (x+3)(x+2)$; Other factor: $(x+2)$
 17. $a^2(b-c) + b^2(c-a) + c^2(a-b) = (a-b)(b-c)(c-a)$ (standard result)
 18. $(997+3)(997-3) = 1000 \times 994 = 994000$
 19. $(x^2+11x+28) = (x+4)(x+7)$; dimensions: $(x+4)$ m and $(x+7)$ m; $x = 5$: $9 \times 12 = 108$ sq. m
 20. $6x^2+17x+12 = (2x+3)(3x+4)$; $x = 2$: $(7)(10) = 70$ sq. m; tile: 7 m $\times$ 10 m (or subunit)

Worksheet 13: Chapter 13 – Introduction to Graphs

Answer Key

Answer Key – Worksheet 13

1. II, IV, III, I 2. (0, 0) 3. 0; 0
4. Teacher Verification Required
5. (a) Line graph; (b) Bar graph or Pie chart
6. Teacher Verification Required
7. (a) $50(5) + 200 = \text{Rs. } 450$; (b) $50(20) + 200 = \text{Rs. } 1200$; (c) 20 books ($C = 1200$)
8. $y = 3x$
9. School= 8.4 h; Sleep= 7.92 h; Play= 3.6 h; Homework= 2.88 h; Other= 1.2 h
10. $d = \sqrt{(3 - (-1))^2 + (4 - 1)^2} = \sqrt{16 + 9} = 5$ units
11. Teacher Verification Required
12. Teacher Verification Required
13. Midpoint= $\left(\frac{4-6}{2}, \frac{-2+8}{2}\right) = (-1, 3)$
14. Mean= $\frac{500 + 700 + 600 + 900 + 800}{5} = \text{Rs. } 700$; Largest growth: Jan to Feb (+Rs. 200)
15. (3, 7): $2(3) + 1 = 7$ ✓; $(-2, -3)$: $2(-2) + 1 = -3$ ✓; Both lie on the line
16. $x + 2 = 3 - x \Rightarrow x = 0.5$; $y = 2.5$; Intersection: (0.5, 2.5); Teacher Verification Required
17. AB= 4, BC= 3, CD= 4, DA= 3; Perimeter= 14 units; Area= $4 \times 3 = 12$ sq. units
18. Slope= $\frac{9-5}{4-2} = 2$; $y - 5 = 2(x - 2) \Rightarrow y = 2x + 1$; When $x = 10$: $y = 21$
19. Teacher Verification Required; Growth= $\frac{5000 - 2000}{2000} \times 100 = 150\%$
20. Plan A: $C = 200 + x$; Plan B: $C = 100 + 2x$; Equal when $200 + x = 100 + 2x \Rightarrow x = 100$ minutes; Teacher Verification Required

Worksheet 14: Chapter 14 – Speed, Distance, and Time**Answer Key****Answer Key – Worksheet 14**

1. 60 km/h 2. 52.5 km 3. $90 \times \frac{5}{18} = 25$ m/s 4. $15 \times \frac{18}{5} = 54$ km/h 5. 6 hours
6. By 8 AM, Bus A has covered 60 km; remaining gap= 390 km; Closing speed= 150 km/h; extra time= 2.6 h; meet at 8AM+2.6h= 10 : 36 AM
7. Total dist= 10 + 15 = 25 km; Total time= 3 h; Avg speed= $\frac{25}{3} \approx 8.33$ km/h
8. Speed= $\frac{200}{10} = 20$ m/s= 72 km/h
9. Let usual time= t ; $\frac{d}{(3/4)v} = t + 20 \Rightarrow \frac{4d}{3v} = t + 20$; but $\frac{d}{v} = t$; so $\frac{4t}{3} = t + 20 \Rightarrow t = 60$ min
10. With wind: $v + w = 600$; Against: $v - w = 500$; $v = 550$ km/h; $w = 50$ km/h
11. In t hours police travel $90t$, thief travels $60(t + \frac{1}{6})$; $90t = 60(t + \frac{1}{6}) \Rightarrow t = \frac{1}{3}$ h= 20

min; Distance= 30 km

12. Meet after $\frac{500}{150} = \frac{10}{3}$ h; from P: $80 \times \frac{10}{3} = 266.7$ km

13. Upstream speed= 10 km/h; downstream= 15 km/h; still water= 12.5 km/h; current= 2.5 km/h

14. $v_{avg} = \frac{2 \times 40 \times 60}{100} = 48$ km/h

15. Journey time= 3 h; must leave by 2PM – 3h = 11 : 00 AM

16. Relative speed= 18 km/h= 5 m/s; Distance= 500 m; Time= 100 s

17. By 10 AM, A has walked 10 km; gap= 34 km; closing speed= 12 km/h; time= $\frac{34}{12} = 2.83$ h; meet at $\approx 12 : 50$ PM

18. Usual time= t ; $d = 8 \left(t + \frac{2}{60} \right) = 10 \left(t - \frac{2}{60} \right)$; $8t + \frac{4}{15} = 10t - \frac{1}{3}$; $2t = \frac{4}{15} + \frac{5}{15} = \frac{9}{15} = 0.6$; $t = 0.3$ h= 18 min; $d = 8 \times \frac{20}{60} = \frac{8}{3} \approx 2.67$ km

19. Avg speed= $\frac{1384}{17} \approx 81.4$ km/h; Running time= 16.5 h; Running speed ≈ 83.9 km/h

20. Time needed= $\frac{3}{4} = 0.75$ h= 45 min; Latest leave= 8 : 00 – 0 : 45 = 7 : 15 AM; If 5 min late (7:20): time left= 40 min; Speed needed= $\frac{3}{2/3} = 4.5$ km/h

Teacher Notes

Chapter 1: A Square and a Cube

Learning Outcomes: Students compute squares and cubes; find square and cube roots; identify perfect squares and cubes; apply these to real-life contexts.

Common Misconceptions: Students often confuse $\sqrt{a+b}$ with $\sqrt{a} + \sqrt{b}$. Emphasise that $\sqrt{9+16} = 5$, not $3+4$. Also, $(-3)^2 = 9$ but $-3^2 = -9$.

Activities: Locker puzzle (which lockers are open after n rounds); staircase of squares; cube-root estimation competitions.

Remediation: Use a table of squares and cubes (1–20) on the wall. Practice prime factorisation before square/cube root.

Chapter 2: Power Play

Learning Outcomes: Students apply laws of exponents; use standard form; evaluate expressions with negative and zero exponents.

Common Misconceptions: $a^m \times a^n \neq a^{mn}$ (add, don't multiply exponents). $a^0 = 1$ surprises students – use the quotient rule to justify. Negative exponents confuse; reinforce $a^{-n} = \frac{1}{a^n}$.

Activities: Exploring powers of 2 (doubling bacteria); reading scientific notation in newspapers; estimation of planetary distances.

Remediation: Create an exponent law reference card. Practice converting to/from standard form daily.

Chapter 3: A Story of Numbers (Rational Numbers)

Learning Outcomes: Students understand rational number properties; perform all four operations; find rational numbers between two given numbers.

Common Misconceptions: Students may invert the wrong fraction when dividing. They may forget to find LCM before adding. Also, the additive inverse of $\frac{-3}{5}$ is $\frac{3}{5}$, not $\frac{-5}{3}$ (that is the multiplicative inverse).

Activities: Number line walks with rational numbers; fraction operation card games; real-life sharing problems.

Remediation: Use the number line constantly. Identify operations step-by-step before calculating.

Chapter 4: Quadrilaterals

Learning Outcomes: Students classify quadrilaterals; apply angle and side properties; solve for unknowns; relate diagonals to quadrilateral types.

Common Misconceptions: Students confuse rhombus and square (both have equal sides, but square has right angles). They may misapply the angle-sum property (only 360° for quadrilaterals, not 180°).

Activities: Quadrilateral family tree poster; geoboard explorations; property hunt in

the classroom.

Remediation: Use a Venn diagram of quadrilateral types. Always draw and label diagrams.

Chapter 5: Number Play

Learning Outcomes: Students use algebra to generalise number patterns; prove divisibility rules; solve number puzzles; understand magic squares.

Common Misconceptions: Students may try to verify properties with just one example rather than proving algebraically. Emphasise: “One example is not a proof.”

Activities: Number tricks for class; magic square creation; investigating the $3n+1$ Collatz conjecture.

Remediation: Always write the algebraic representation first (e.g. $10a + b$ for a 2-digit number) before exploring properties.

Chapter 6: Algebraic Identities

Learning Outcomes: Students expand and factorise using identities; compute products mentally; apply identities to real-life problems.

Common Misconceptions: $(a + b)^2 \neq a^2 + b^2$ (missing the $2ab$ term). Students must memorise: always include the middle term.

Activities: Geometric proof of $(a + b)^2$ using a square diagram; mental calculation competitions using identities.

Remediation: Derive identities from first principles (expand and verify). Post identities on the classroom wall.

Chapter 7: Proportional Reasoning – 1

Learning Outcomes: Students solve ratio, proportion, profit/loss, SI, and CI problems; understand percentage applications.

Common Misconceptions: CI is not “more SI”; it is interest on interest. Students may confuse profit% on CP vs SP. Clarify: profit% is always calculated on CP unless stated otherwise.

Activities: Bank visit/simulation; market role-play (buying, selling, discounts); EMI calculations.

Remediation: Always identify what the percentage is “of” before calculating. Draw a flowchart for CI.

Chapter 8: Playing with Constructions

Learning Outcomes: Students construct quadrilaterals using compass and ruler given various combinations of measurements.

Common Misconceptions: Students may not understand why 5 measurements are needed (a quadrilateral has more degrees of freedom than a triangle). They may try to construct from insufficient data.

Activities: Art using geometric constructions; architectural design on paper; Islamic geometric pattern creation.

Remediation: Always check: do I have enough independent pieces of information? Use a construction checklist.

Chapter 9: Mensuration

Learning Outcomes: Students calculate surface area and volume of cubes, cuboids, and cylinders; solve composite solid problems.

Common Misconceptions: Using diameter instead of radius in circle/cylinder formulas is the most common error. Also, TSA includes top and bottom; LSA (curved SA) does not.

Activities: Measuring and calculating SA/volume of classroom objects (boxes, cans, tins); estimating water in tanks.

Remediation: Write formulas with labels. Always check units at the end (sq. m vs cubic m).

Chapter 10: Statistics and Probability

Learning Outcomes: Students compute mean, median, mode for grouped and ungrouped data; distinguish experimental from theoretical probability; compute probability of simple events.

Common Misconceptions: Students may confuse the sum of probabilities (always 1 for all mutually exclusive outcomes) with individual probabilities. For grouped data, they may not use midpoints for mean.

Activities: Coin toss experiment (100 trials); die roll frequency table; class survey for data handling.

Remediation: Emphasise sorting data before finding median. Use the formula $P(A) = \frac{\text{favourable}}{\text{total}}$ consistently.

Chapter 11: Linear Equations

Learning Outcomes: Students solve linear equations with variables on both sides, with fractions, and apply to word problems.

Common Misconceptions: Students may move terms incorrectly (changing sign when transposing is the most common error). They may also forget to multiply all terms when clearing fractions.

Activities: Balance scale model for equations; age-puzzle competitions; translating real-life sentences to equations.

Remediation: Always verify solutions by substituting back. Teach “BODMAS after transposing” clearly.

Chapter 12: Factorisation

Learning Outcomes: Students factorise using common factors, grouping, and identities; divide polynomials by monomials and binomials.

Common Misconceptions: Students may not factorise completely (e.g. stopping at $2(a^2 - 25)$ instead of $2(a+5)(a-5)$). Remind them to always check if further factorisation is possible.

Activities: Factorisation puzzles; product/sum pair games; area model for factorisation.

Remediation: Always ask: “Is there a common factor first?” Then check for difference of squares or perfect square trinomials.

Chapter 13: Introduction to Graphs

Learning Outcomes: Students plot points in all four quadrants; draw and interpret line graphs; identify linear relationships; use graphs to solve problems.

Common Misconceptions: Students often mix up (x, y) order – always “across first, then up/down.” They may draw bar graphs when a line graph is more appropriate for continuous data.

Activities: Real-data graphing (temperature, sales, population); distance–time graph activities; graphing calculator or GeoGebra exploration.

Remediation: Practice plotting points daily as a warm-up activity. Relate each axis to a real quantity.

Chapter 14: Proportional Reasoning – 2 (Speed, Distance, Time)

Learning Outcomes: Students apply $S = D/T$; convert units; solve relative speed, average speed, and meeting-point problems.

Common Misconceptions: Average speed is NOT the arithmetic mean of speeds (unless time intervals are equal). Students may forget to align units (km vs m, hours vs seconds).

Activities: Stopwatch speed experiments on the playground; reading train timetables; planning a road trip.

Remediation: Always write $D = S \times T$ and identify what is known before solving. Draw a timeline for meeting-point problems.

End of Grade 8 Mathematics Worksheet Book

Based on NCERT *Ganita Prakash* Parts 1 & 2 – 14 Chapters, 14 Worksheets

“Without mathematics, there’s nothing you can do. Everything around you is mathematics. Everything around you is numbers.” — Shakuntala Devi
